

ABDULLAH KHAWAJA

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TECHNICAL ARSENAL

GenAI & LLMs:	Unsloth, LangChain, Ollama, Claude Code, OpenAI, Cursor, Gemini-CLI, Fine-tuning (Gemma/Llama).
Machine Learning:	TensorFlow, Keras, PyTorch, Scikit-Learn, XGBoost, Vertex AI, BigQuery, Recommender Systems (Wide & Deep, LTR).
Development:	Python, Java, C++, React, TypeScript, Electron, SQL, Docker, Git.
Cloud & Tools:	Alibaba Cloud, AWS, GCP, Vercel, Postman, Hetzner, Tableau.

WORK EXPERIENCE

Machine Learning Engineer | DPD Group

Nov 2023 – Present

- **Chatbot Fine-Tuning:** Fine-tuned **Gemma** for the DPD chatbot using internal support conversations, tool calls, and databooks. The work focused on lowering serving cost while keeping the chatbot answers grounded in the operational context people already relied on.
- **Customer Retention (Churn Prediction):** Architected a 6-model meta-ensemble (including Dual LightGBMs, WaveNet, Logistic Hazard, and a Hybrid XGBoost-Transformer) that achieved the **highest predictive performance across DPD Group**. Accurately identified **64% of flagged customers** as true churners and captured **67% of total churn cases**, with **savings of \$650K recorded so far**.
- **Volume Forecasting:** Developed a deep learning model (Mixture of Experts, RNN, WaveNet) surpassing a 12-year human benchmark by **+8%**. Replaced fixed lags with **event-based lags** (Amazon Prime Day, Bank Holidays) to capture seasonality accurately, delivering consistent daily accuracy for 80+ hubs.

Data Scientist (Contract) | Dastgyr (Remote)

Jun 2023 – Nov 2023

- **Recommender Systems:** Implemented SWING collaborative filtering from scratch for the "Top For You" module, driving a **+1.44% uplift in basket size** and **+22% increase in average order value**.
- **Strategy:** Defined the AI roadmap (sales forecasting, OOS reduction) and fostered a culture of experimentation and documentation.

Senior Data Scientist – Recommendations | Daraz (Alibaba Group)

Oct 2020 – Feb 2023

- **Rank Stage:** Replaced logistic regression with **Wide & Deep** learning models, achieving **+3% CTR uplift** via advanced feature engineering and DL infrastructure integration.
- **Model Optimization:** Reduced feature space by 50% using feature importance, slashing backup recommendation compute time from **3h to 1h 25m** with no metric degradation (handling 60% of all recommendations).
- **Learning to Rank:** Introduced pointwise LTR to handle discount price/ratio trade-offs, driving a **+35% increase in conversion rate (CR)**.
- **Leadership:** Led 50+ A/B tests across 5 countries; top-rated performer for 2 consecutive years.

PROJECTS

Sort Moments | [GitHub](#)

- **Tech:** Python, PyQt6, RetinaFace, Buffalo Large (Embeddings).
- **Description:** A privacy-first desktop application engineered to tackle the chaos of large-scale event photography (4,000+ images). It runs a fully local ML pipeline (RetinaFace + InsightFace) using DirectML (blazing fast inference across all GPUs) to detect and cluster faces with high accuracy, automatically organizing solo and group shots by person without data ever leaving the device.
- **Features:** Offline processing, rapid image size reduction, high-precision face clustering, and a streamlined workflow for manual refinement and deduplication.

WeaveSkip

- **Tech:** Next.js, TypeScript, Prisma, PostgreSQL, Redis/BullMQ, NextAuth, OAuth, S3, Stripe, Zod.
- **Description:** A creator distribution OS that turns one article, RSS feed, or manual source into platform-native drafts for LinkedIn, X, Reddit, Facebook, Threads, and Substack Notes.
- **Features:** Source ingestion, intent/keyword/subreddit intelligence, BYOK LLM config with encrypted keys, human review, guardrails, queue-based scheduling/publishing, and Medium crosspost support.

BookmarkFeed

- **Tech:** Electron, React/Vite, FastAPI, Playwright, SQLite, ChromaDB, Sentence Transformers.
- **Description:** A privacy-first desktop app that aggregates saved posts/bookmarks from X/Twitter, Reddit, LinkedIn, Instagram, Facebook, and Threads into a local searchable archive.
- **Features:** Local media capture, cross-platform fetchers, semantic search, SQLite metadata, Chroma vectors, and a Python sidecar bundled with Electron.

Hugging Face Model Releases

- **Tech:** Gemma, Qwen, LoRA/fine-tuning, ablation, distillation, GGUF quantization.
- **Description:** Public model releases exploring small-model adaptation, ablation experiments, Opus-distilled outputs, and deployment-ready GGUF packaging.
- **Focus:** Hands-on LLM training, model surgery, quantization, and open-source distribution across multiple experimental releases.

Navigating Noise | [HackerNoon](#)

- **Focus:** Technical writing on ML systems, AI tools, benchmarks, interviews, and product lessons for engineers and builders.
- **Description:** Converts noisy AI developments into practical mental models: what to follow, how to prepare, and how to judge tools that claim to change workflows.

EDUCATION

Master of Science in Data Science (Distinction) | University of Essex, UK

Oct 2022 – Oct 2023

Bachelor of Science in Computer Science | FAST-NU, Pakistan

Aug 2016 – Sep 2020